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Toxicological impact of diafenthiuron (50% WP) on water quality, physiology, behaviour and histopathology of the freshwater catfish *Clarias batrachus*

Neetu Singh*, Ravindra Kumar Singh & Sanjeev Kumar

P.G. Department of Zoology, Jai Prakash University, Chapra, Bihar, India

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Abstract- Diafenthiuron (50% WP) is a widely applied pyrethroid pesticide that frequently contaminates freshwater ecosystems through agricultural runoff. The present investigation evaluates its toxicological impact on water quality parameters, acute toxicity, oxygen consumption, behavioural and morphological responses, biochemical and haematological alterations, and histopathological changes in the freshwater air-breathing catfish *Clarias batrachus*. Laboratory-based bioassays were conducted using sub-lethal and lethal concentrations of Diafenthiuron under controlled conditions. Exposure resulted in significant deterioration of water quality, reduced dissolved oxygen, altered respiratory metabolism, abnormal behaviour, morphological deformities, metabolic imbalance, haematological disturbances, and severe liver tissue damage. The study clearly demonstrates that Diafenthiuron poses a serious ecological threat to freshwater fishes even at low concentrations and prolonged exposure durations.

Keywords: Diafenthiuron, freshwater toxicity, *Clarias batrachus*, LC_{50}, behavioural stress, histopathology

INTRODUCTION

The increasing use of synthetic pesticides in modern agriculture has raised serious concerns regarding their persistence and toxicity in aquatic ecosystems. Runoff from agricultural fields, improper disposal, and accidental spills introduce these chemicals into freshwater bodies, where they adversely affect non-target organisms. Fish are particularly vulnerable to pesticide contamination due to their direct contact with polluted water and their ability to bioaccumulate toxic substances.[1]

Diafenthiuron (50% WP) is a pyrethroid pesticide extensively used for pest control in agricultural practices. Although considered effective against target pests, its impact on aquatic fauna remains inadequately documented. Pyrethroids are known to interfere with nervous transmission, respiratory metabolism, and cellular integrity in fishes. Behavioural and morphological responses are among the earliest indicators of pesticide stress and are often accompanied by biochemical, haematological, and histopathological changes.[2]

The freshwater catfish *Clarias batrachus* is an economically important species widely consumed in India. Being an air-breathing fish, it is frequently exposed to pollutants in stagnant and semi-stagnant water bodies. Alterations in its physiological and behavioural parameters can therefore serve as reliable indicators of environmental toxicity. The present study was designed to evaluate the acute and sub-lethal effects of Diafenthiuron (50% WP)

*Corresponding author :
Phone : 7903494990
E-mail : nitu3146@gmail.com

on *C. batrachus*, with special emphasis on water quality, mortality, respiration, behaviour, metabolism, blood parameters, and tissue damage.

MATERIALS & METHODS

Healthy specimens of the freshwater catfish *Clarias batrachus* (6–8 ± 0.5 cm in length), representing both sexes, were procured from the local fish market of Chapra and acclimatised to laboratory conditions for one week prior to experimentation. During the acclimatisation period, the fishes were fed daily with a diet consisting of groundnut cake and rice bran. Any batch showing mortality greater than 5% during acclimatisation was excluded from the study to ensure uniform health status of the experimental animals. All experimental procedures were carried out following standard guidelines prescribed by the American Public Health Association.[3] The water quality maintained throughout the experimental period included a temperature of 28 ± 2°C, pH of 8.2, dissolved oxygen levels ranging from 8 to 10 mg L^{-1}, and total hardness of approximately 320 mg L^{-1}.

Acute toxicity experiments were conducted using graded concentrations of Diafenthiuron (50% WP). Fish mortality was recorded at 24, 48, 72, and 96 hours of exposure, and LC_{50} values were calculated using probit analysis.[4] Physico-chemical parameters of water, including temperature, pH, dissolved oxygen, electrical conductivity, and total hardness, were monitored throughout the experimental period. Oxygen consumption was measured to assess alterations in respiratory metabolism.

Behavioural and morphological responses were visually observed and systematically recorded at regular intervals. Biochemical analyses included the estimation of plasma glucose, plasma protein, liver glycogen, and muscle glycogen to evaluate metabolic stress. Haematological parameters were analysed to assess physiological responses, while histopathological examination of liver tissues was performed using standard fixation, sectioning, and staining techniques to determine cellular-level damage.

RESULTS

Influence of Diafenthiuron on Water Quality Parameters

Diafenthiuron exposure resulted in noticeable changes in water quality parameters throughout the study period (Table 1). A gradual increase in temperature, pH, and electrical conductivity was observed with rising pesticide concentration. Conversely, dissolved oxygen and total hardness declined significantly. The reduction in dissolved oxygen suggests increased metabolic demand and impaired respiratory efficiency in exposed fish. The pesticide concentration was maintained below 1.00 mg L^{-1} to evaluate its influence on water quality without causing immediate lethality.

Table 1: Influence of Diafenthiuron (50% WP) on water quality parameters

Conc (mg/l)	Temperature (°C)	pH	DO (mg/l)	EC (µs/ Cm^{-1})	Total Hardness (mg/l)
00	24.56	7.52	8.26±0.74	227	212
0.54	24.85	7.61	8.03±0.73	242	205
0.62	24.98	7.74	7.81±0.71	257	197
0.76	25.13	7.82	7.62±0.58	271	189
0.91	25.27	7.91	7.43±0.57	286	181
1.06	25.41	8.03	7.21±0.60	302	173

Acute Toxicity and Mortality Response

Fish mortality increased progressively with increasing Diafenthiuron concentration and exposure duration (Table 2). No mortality was observed in the control group, indicating suitable experimental conditions. The LC_{50} value calculated through probit analysis was 5.28mg L^{-1}, with lower and upper confidence limits of 1.92 and 7.56mg L^{-1}, respectively. The estimated safe and dischargeable limits were 1.27 mg L^{-1} and 1.96 mg L^{-1} for long-term exposure.

A clear decline in LC_{50} values with increasing exposure time confirmed that Diafenthiuron becomes more toxic with prolonged exposure, even at lower concentrations.

Table 2: Fish mortality in number and % mortality of *Clarias batrachus* at Diafenthiuron 50 WP treatments during the experimental period

Concentration (mg/l)	24h	48 h	72 h	96 h	Log Conc (Log10)	Probit
Control	00	00	00		-	-
1.92	00	00	2 (10)	3 (15)	0.26840	1.5221
3.05	00	2 (10)	3 (15)	5 (25)	0.39555	2.1987
4.19	2 (10)	3 (15)	5 (25)	9 (45)	0.50585	2.9176
5.31	3 (15)	5 (25)	8 (40)	12 (60)	0.72585	3.4868
6.43	4 (20)	9 (45)	12 (60)	16 (80)	0.80156	4.1734
7.56	6 (30)	10 (50)	18 (90)	20 (100)	1.00000	-

Effect on Oxygen Consumption

Oxygen consumption in *C. batrachus* decreased significantly under both sub-lethal and lethal Diafenthiuron exposure (Table 3). The decline was gradual but consistent from 0 to 96 hours, indicating suppression of respiratory metabolism. Reduced oxygen uptake reflects gill damage, excessive mucus secretion, and impaired physiological regulation of respiration.[5]

Table 3: Oxygen consumption in *Clarias batrachus* exposed to sub-lethal and lethal Diafenthiuron (50 WP) exposure.

Sl. No.	Exposure Time (hr)	Control (mg/l)	Sub-lethal (mg/l)	Lethal (mg/l)
1	0	8.62	8.13	7.76
2	24	8.21	7.78	7.38
3	48	7.82	7.36	7.09
4	72	7.43	6.97	6.68
5	96	7.07	6.43	6.16

Behavioural Alterations

Behavioural changes were among the most prominent indicators of Diafenthiuron toxicity (Table 4). During early exposure, fish exhibited hyperactivity, frequent surfacing, and increased opercular movement, suggesting acute respiratory stress. As exposure progressed, opercular movements slowed, swimming activity became sluggish, and fish displayed loss of equilibrium.[5,6]

Erratic swimming, wall hitting, and eventual paralysis were commonly observed at higher concentrations. The lethal dose induced abnormal behaviour within one hour, including paralysed fins and irregular opercular action, ultimately leading to mortality.

Table 4: *Clarias batrachus* behavior after different Diafenthiuron exposure (L=Less; M=More; F=Fast; N=Normal; S=Slow; P=Present; A=Absent; C=Control; C1=LC_{10}; C2=LC_{25}; C3=LC_{50}).

	24 h			48 h				72 h		96 h		
	C1	C2	C3	C1	C2	C3	C1	C2	C3	C1	C2	C3
Erratic swimming	A	A	A	A	A	L	A	M	M	L	M	M
Air at surface	A	A	L	A	A	L	A	M	M	L	L	M
Operculam movement	N	N	N	N	N	N	N	F	F	N	F	S
Loss of balance	A	A	A	A	A	A	A	A	L	A	L	L
Hitting against wall	A	A	P	P	A	A	A	P	P	P	P	P
Sluggishness	A	A	A	A	A	A	A	P	P	P	P	P

Morphological Changes

Diafenthiuron caused visible morphological deformities that intensified with exposure duration and concentration (Table 5). These included loosening of scales, redness of eyes, excessive mucus secretion, gill bleeding, abdominal swelling, and dark pigmentation patches. At lethal concentrations, fish showed ruptured fin rays, haemorrhagic gills, swollen abdomen, and discoloured body surface. These changes reflect severe physiological stress and tissue damage.

Biochemical Responses

Exposure to Diafenthiuron resulted in a significant increase in plasma glucose levels, indicating stress-induced hyperglycaemia (Table 6). Plasma protein levels declined steadily, reflecting impaired protein synthesis or enhanced protein degradation. Liver and muscle glycogen levels increased, suggesting altered carbohydrate metabolism and mobilisation of energy reserves to cope with toxic stress.

Table 5: Morphological changes of different Diafenthiuron exposure on *Clarias batrachus* (L=Less; M=More; F=Fast; N=Normal; S=Slow; P=Present; A=Abscent; C1=LC_{10}; C2=LC_{25}; C3=LC_{50}).

Morphological changes	24 h			48 h				72 h		96		
	C1	C2	C3	C1	C2	C3	C1	C2	C3	C1	C2	C3
Loosening of scale	A	A	A	A	A	A	A	A	A	A	P	P
Redness in Eyes	A	A	A	A	A	A	A	A	P	A	P	P
Profuse mucous loss	A	A	A	A	A	A	A	P	P	A	P	P
Blooding from Gills	A	A	A	A	A	A	A	A	P	A	P	P
Belly balloning	A	A	A	A	A	A	A	A	P	A	A	P
Dark patches in abdomen	A	A	A	A	A	A	A	A	P	A	P	P

Table 6: Biochemical parameters in fish due to Diafenthiuron (50 WP) exposure.

Biochemical parameters	Fenvalerate exposure in hours				
	Control	24 hr	48 hr	72 hr	96 hr
Plasma glucose (mg/l)	117	123	126	131	137
Plasma protein (mg/l)	16	15	13	12	10
Liver glycogen (µg/ml)	132	134	137	141	146
Muscle glycogen (µg/ml)	148	150	153	156	160

Haematological Alterations

Haematological parameters showed non-significant variation after 24 hours of exposure but exhibited pronounced changes after 48 to 96 hours (Table 7). Red blood cell counts and haemoglobin levels declined progressively, indicating reduced oxygen-carrying capacity. White blood cell counts initially increased, suggesting an immune response, but later declined under prolonged exposure, indicating physiological exhaustion.

Table 7: Haematological changes in fish due to Diafenthiuron (50 WP) exposure.

Biochemical parameters	Diafenthiuron exposure in hours				
	Control	24 hr	48 hr	72 hr	96
RBC (10^6/10mm^3)	0.54	0.53	0.51	0.48	0.44
WBC (10^6/10mm^3)	30	31	33	35	38
MCV	95	96	99	101	104
MCH	40	41	42	44	47
MCHC	32	33	34	36	39

Histopathological Changes

Histological examination of liver tissues revealed severe pathological alterations. At lower concentrations, cytoplasmic vacuolation and vascular dilation were

observed. Moderate concentrations caused erythrocyte congestion, sinusoidal dilation, and necrosis. At higher concentrations and prolonged exposure, extensive necrosis and hyperaemia were evident. The highest concentration resulted in widespread tissue destruction, confirming the hepatotoxic nature of Diafenthiuron.

DISCUSSION

The findings of the present investigation clearly demonstrate that Diafenthiuron (50% WP) exerts severe toxic effects on *Clarias batrachus*. Alterations in water quality, particularly dissolved oxygen depletion, contribute to respiratory stress. The progressive decline in LC_{50} values with exposure duration highlights the cumulative toxicity of the pesticide. Behavioural abnormalities such as erratic swimming, loss of balance, and surfacing indicate nervous system impairment, likely due to disrupted nerve impulse transmission[7] and inhibition of acetylcholine activity. Excessive mucus secretion and gill damage further impair respiratory function.[8,9] Biochemical and haematological alterations reflect metabolic imbalance and physiological stress[10], while histopathological damage to the liver confirms cellular-level toxicity. Although Diafenthiuron caused less severe effects compared to Diazinon, its impact remains ecologically significant.

CONCLUSION

The present study demonstrates that Diafenthiuron (50% WP) causes significant toxic effects in the freshwater catfish *Clarias batrachus* even at low concentrations. Exposure to the pesticide altered water quality parameters, particularly dissolved oxygen and electrical conductivity, leading to respiratory and physiological stress in the fish. Acute toxicity tests showed concentration- and time-dependent mortality, with decreasing LC_{50} values indicating increased toxicity with prolonged exposure.

Behavioural and morphological abnormalities such as erratic swimming, loss of equilibrium, excessive mucus secretion, and gill damage were prominent indicators of pesticide stress. Biochemical and haematological changes further confirmed metabolic imbalance and reduced oxygen-carrying capacity. Histopathological damage to liver tissues, including vacuolation and necrosis, highlighted the hepatotoxic nature of Diafenthiuron. Overall, the findings indicate that Diafenthiuron poses a potential ecological risk to freshwater fishes and may adversely affect fish health and survival under continuous exposure. Careful regulation of its use and routine monitoring of aquatic environments is essential to minimise its impact on freshwater ecosystems.

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